

CAP-D'ESPOIR, QC, Canada

WMO: 714290

Lat: 48.419N

Lon: 64.317W

Elev: 15

StdP: 101.14

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.2	-18.6	-27.7	0.3	-20.5	-25.0	0.4	-18.4	18.2	-2.3	15.9	-1.8	4.5	270	0.737

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.7	21.2	17.6	20.1	17.3	19.1	16.8	18.9	20.2	18.1	19.4	17.3	18.5	4.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	13.2	19.6	17.5	12.6	18.9	16.8	12.0	18.2	53.8	20.3	51.4	19.4	49.0	18.6	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.5	12.5	10.8	-23.9	25.1	2.4	2.2	-25.6	26.6	-27.1	27.9	-28.4	29.1	-30.2	30.7		
			DB	WB	-24.3	20.5	2.2	0.9	-25.9	21.1	-27.2	21.7	-28.4	22.2	-30.0	22.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.7	-8.9	-8.3	-4.4	0.8	6.3	11.6	15.7	15.9	12.1	6.4	1.1	-4.8
	DBStd	9.36	5.37	5.15	4.33	2.77	2.72	2.87	2.19	2.03	2.94	3.10	3.71	4.54
	HDD10.0	2813	585	513	447	274	119	16	1	0	13	117	268	460
	HDD18.3	5354	843	747	706	524	372	203	86	79	189	368	518	718
	CDD10.0	508	0	0	0	0	5	62	176	183	75	7	0	0
	CDD18.3	8	0	0	0	0	0	0	4	4	1	0	0	0
	CDH23.3	6	0	0	0	0	0	1	2	2	1	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	4.8	5.1	5.1	5.4	5.0	4.6	4.1	3.6	3.9	4.5	5.0	5.2	5.5
Precipitation	PrecAvg	1125	80	78	86	78	94	97	99	97	99	115	97	100
	PrecMax	1455	130	151	176	154	175	191	157	158	163	231	192	215
	PrecMin	904	27	10	17	27	43	39	46	38	44	37	33	31
	PrecStd	128	31	34	39	38	38	34	28	33	33	43	34	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.1	3.1	5.1	9.6	16.9	21.3	23.2	23.2	21.0	16.2	10.1	5.3
		MCWB	3.2	1.9	1.9	4.8	11.5	15.3	18.0	18.2	16.7	12.1	8.4	4.2
	2%	DB	2.0	1.4	2.9	7.0	13.8	18.8	21.2	21.2	18.7	13.5	8.6	3.6
		MCWB	1.4	0.6	0.9	3.9	10.3	15.3	17.8	18.0	16.3	11.4	7.5	2.9
	5%	DB	0.5	0.1	1.6	5.5	11.9	17.1	20.1	20.1	17.3	12.1	7.4	2.4
		MCWB	-0.1	-0.4	0.3	3.2	9.1	14.5	17.4	17.6	15.4	10.7	6.4	1.6
	10%	DB	-1.1	-1.1	0.7	4.3	10.3	15.7	19.1	19.2	16.3	10.9	6.1	1.4
		MCWB	-1.8	-1.7	-0.3	2.5	8.0	13.5	16.7	17.2	14.6	9.5	5.0	0.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	2.2	3.0	6.2	13.0	18.0	20.2	20.1	18.6	13.7	9.2	4.7
		MCDB	3.9	2.5	4.1	8.0	15.1	19.3	21.7	21.6	19.6	15.1	9.4	4.8
	2%	WB	1.4	0.7	1.5	4.6	11.1	16.3	18.9	19.0	17.0	12.0	7.7	3.0
		MCDB	1.8	1.2	2.3	5.7	13.0	17.7	20.3	20.2	17.9	12.9	8.2	3.3
	5%	WB	-0.2	-0.4	0.6	3.6	9.7	15.1	18.0	18.3	15.9	10.9	6.3	1.7
		MCDB	0.2	0.0	1.4	4.7	11.3	16.6	19.3	19.5	16.8	11.9	6.9	2.2
	10%	WB	-1.9	-1.6	-0.2	2.7	8.4	13.9	17.2	17.6	14.9	9.7	5.1	0.7
		MCDB	-1.4	-1.2	0.6	3.9	9.9	15.4	18.5	18.8	16.1	10.8	5.7	1.2
Mean Daily Temperature Range	5% DB	MDBR	7.5	6.9	5.8	4.8	5.5	5.5	5.4	5.7	6.3	5.6	5.2	6.1
		MCDBR	8.8	6.0	6.0	6.5	8.3	7.7	7.0	6.9	7.1	6.9	5.7	5.7
		MCWBR	9.5	5.9	5.1	4.9	5.8	5.4	5.0	5.0	5.6	5.7	5.7	6.0
	5% WB	MCDBR	9.3	5.7	5.6	5.9	7.2	6.7	6.0	5.6	6.2	6.4	5.7	5.8
		MCWBR	9.6	5.8	5.2	4.8	5.7	5.4	4.8	4.6	5.3	5.9	5.8	6.2
Clear-Sky Solar Irradiance	taub		0.278	0.282	0.315	0.349	0.366	0.384	0.414	0.388	0.360	0.340	0.309	0.287
	taud		2.380	2.300	2.300	2.389	2.423	2.374	2.338	2.404	2.467	2.494	2.495	2.422
	Ebn at Noon		798	878	900	902	896	878	843	852	845	804	758	738
	Edh at Noon		67	90	107	110	111	117	120	107	92	76	61	58
All-Sky Solar Radiation	RadAvg		1.21	2.04	3.31	4.39	5.16	5.59	5.40	4.76	3.64	2.25	1.18	0.87
	RadStd		0.11	0.20	0.27	0.37	0.41	0.45	0.47	0.30	0.29	0.17	0.14	0.08

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)	N/A	N/A	N/A	+1.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page